

Mohammad Nusairat

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EDUCATION

University of Illinois at Chicago

Chicago, IL

B.S. in Data Science with a Concentration in Computer Science, GPA: 3.50

May 2025

- Related Coursework: Data Structures & Algorithms, Advanced Data Structure Practicum, Intro to Machine Learning, Software Design, Database Systems, Optimization for Analytics, Business Project Management

SKILLS

Programming Languages: Python, C++, C, C#, SQL, Java, Javascript, TypeScript, R, SAS

Data Science Libraries: Pandas, NumPy, TensorFlow, PyTorch, Scikit-learn, Matplotlib, Seaborn

Frameworks & Technologies: Flask, React, Tailwind CSS, MongoDB, PostgreSQL

Development Tools: Git, VS Code, Jupyter Notebook, Anaconda, Microsoft Office Tools

EXPERIENCE

Comprehension, Collaboration and Creativity Lab

University of Illinois at Chicago, IL

Undergraduate Research Assistant

Sept 2024 - May 2025

- Research cognitive patterns in **algorithmic** problem-solving for a study on “Aha! moments” in programming
- Analyze participant interviews, **annotated** insight-related data, **co-authoring** findings **presented** at MPA 2025

Outlier | Part-Time

Remote

Software Developer - AI Trainer

July 2024 - Present

- Evaluate the quality of AI-generated Python/C++ code by developing **robust test cases** to train the model
- Enhance the accuracy of LLM’s code through **data ingestion** by solving complex algorithmic problems

Headstarter AI

Remote

Software Engineering Fellow

July 2024 - August 2024

- Built 5 **AI apps** & APIs using NextJs, OpenAI, Pinecone, StripeAPI with 98% accuracy as seen by 1000 users
- Developed projects from design to deployment **leading 4 engineering fellows** using **MVC** design patterns
- Coached by Amazon, Bloomberg, and Capital One engineers on **Agile**, **CI/CD**, **Git** and **microservice** patterns

PROJECTS

Obesity Risk Predictor

November 2024 - Present

- Designed a **machine learning pipeline** in **Python** to predict obesity risk from lifestyle and demographic factors
- Engineered features from dataset of **20,000 records**, implementing Logistic Regression to identify key features
- Implemented Naive Bayes, SVC, and Ensemble Method using Support Vector, Random Forest, and Gradient Boosting Classifier achieving **98% accuracy**, allowing for high interpretability and actionable health insights
- Currently expanding into a **Dockerized** full-stack app with **JWT**, **PostgreSQL** database, and **AWS** deployment

Campus Meeting Point Application

January 2025 - Present

- Engineered a **full-stack web application** for UIC’s undergraduate campus, to be deployed to the student body
- Built using **Flask** and **React**, the app uses **XML parsing of OpenStreetMap data** to calculate the optimal meeting point between any number of users on campus, filtered by user-selected building utilities and features
- Developed scalable **REST API** endpoints supporting building autocomplete, tag-based filtering, and geospatial pathfinding using **Dijkstra’s algorithm** and **Geodesic Fermat point logic**
- Engineered fault-tolerant location logic and dynamic frontend components for search, filters, and ETA summaries, enabling real-time user coordination with location-aware filtering via **NetworkX** and **geopy**

Flight Delay Prediction

August 2024 - December 2024

- Developed a **machine learning pipeline** in **Python** to predict flight delays from O’Hare International Airport
- Trained models Gradient Boosting, Random Forest, Logistic Regression, and KNN achieving **81% accuracy**
- Published **visualizations** to offer actionable insights to optimize airline scheduling to minimize delays

PUBLICATIONS

Rahman, J., Virruso, A., Nusairat, M., Khan, R., Miller, T. S., & Wiley, J. (2025, April 10-12). Examining the Aha! experience in computer programming. Poster presented at the 97th Annual Meeting of the Midwestern Psychological Association, Chicago, IL